

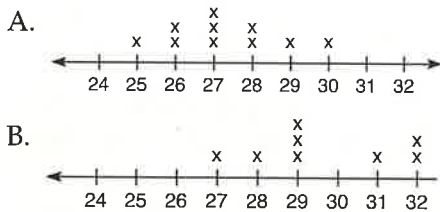
Practice 5

For questions 1 and 2, choose the line plot that matches each frequency table. Use choices A–D below for both questions.

1. Zoologists have counted the number of stripes on certain zebras in a zoo. The set of data from these observations is represented by the following frequency table:

Number of Stripes per Side	Number of Zebras with That Many Stripes
24	I
25	0
26	II
27	0
28	II
29	0
30	II
31	0
32	I

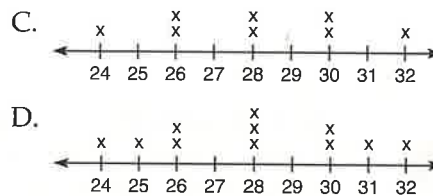
Which line plot represents this data?



2. At a certain elementary school, classes may have different numbers of students. The following frequency table represents how many students are in the classes this year:

Number of Students	Number of Classes with That Many Students
24	0
25	0
26	0
27	I
28	I
29	III
30	0
31	I
32	II

Which line plot represents this data?

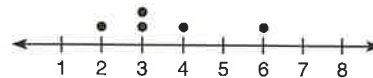


Questions 3 and 4 are based on the following information.

Juana runs a community garden. She is interested in how many different types of vegetables garden members are planting. She collects the following data:

Garden Member	Types of Vegetables Planted
Alberta	4
Paul	3
Eileen	6
Donato	2
Vijay	3
Janelle	2
Zara	1
Wahib	8
Bill	4
Yeo	3

3. The line plot below is based on the information in the table. Each point represents one of the members of the community garden. Place as many additional points as necessary to complete the line plot.



4. Which value (number of types of vegetables) has the highest frequency in the data?
- A. 2
B. 3
C. 4
D. 8

Answers and explanations begin on page 668.

Lesson 6: Histograms

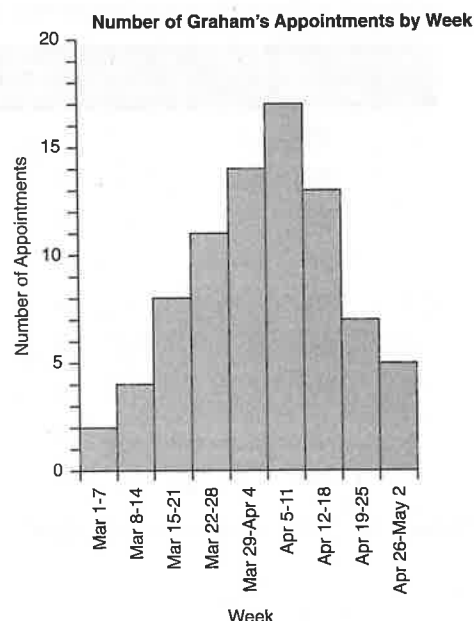
Like line plots, **histograms** display **frequencies**, but they do so in a very different way. Consider the following example:

Key Ideas

- A histogram is another form of data display based on groups and counts within a group.
- Groups are plotted on the horizontal axis, and frequencies are plotted on the vertical axis.

Example 1: Graham is a veterinarian. Last year he decided to track client appointments for spring shots. The following table and histogram represent his results:

Week of	Number of Appointments
March 1–7	2
March 8–14	4
March 15–21	8
March 22–28	11
March 29–April 4	14
April 5–11	17
April 12–18	13
April 19–25	7
April 26–May 2	5



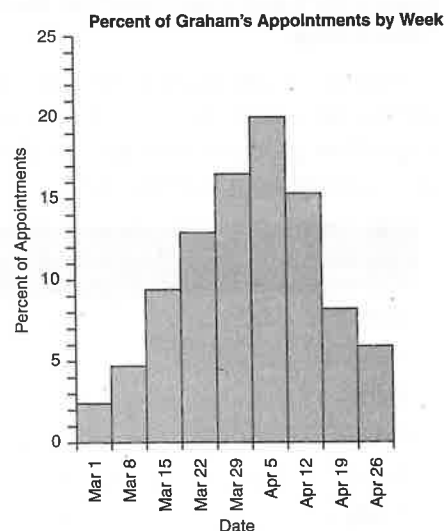
ON THE GED® TEST

Some multiple-choice questions may ask you to identify graphs that represent different types of data collection. You may be given some data and asked which graph represents that set of data.

The **histogram** above on the right has two axes: a **vertical axis** (or **y-axis**) representing numbers of appointments and a **horizontal axis** (also called an **x-axis**) that has increments representing the weeks. The **area** of each bar represents the number of appointments for that week. Thus, you can see not only how many appointments were in any given week but also how the **frequency** of appointments changed over the weeks in the chart.

Finally, histograms can also be used to show percentages. If we alter the histogram about Graham's appointments so that each bar represents the **percent** of total appointments each week, it would look like the one on the right.

Because the percentages add up to 100%, the areas of all the bars add up to 100.



Example 2: What percent of appointments for spring shots occurred during the first two weeks of March?

You can figure this out by adding the percentages for those weeks (you can approximate based on the histogram): for March 1–7, about 2%; for March 8–14, about 5%. Approximately 7% of appointments for spring shots occurred during those two weeks.

Practice 6

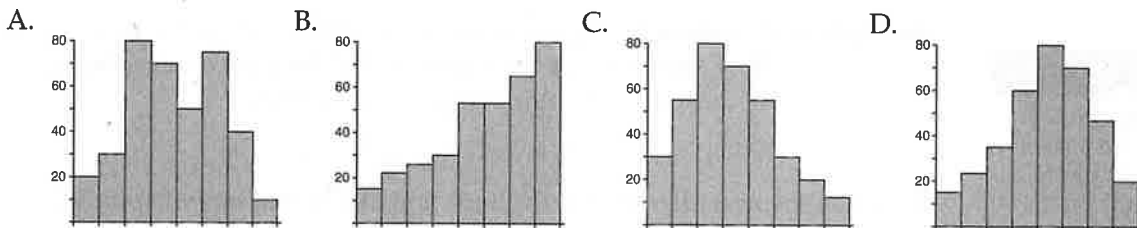
A. Choose the histogram that matches each frequency table. Use choices A–D below for both questions 1 and 2.

1. Josefina runs a retail business. She posted a coupon to a social media site from 9:00 a.m. to 5:00 p.m. The table shows how many people downloaded or printed the coupon.

Hour Beginning With:	Number of People Who Downloaded or Printed Coupon:
9:00 a.m.	30
10:00 a.m.	56
11:00 a.m.	80
12:00 p.m.	71
1:00 p.m.	56
2:00 p.m.	31
3:00 p.m.	22
4:00 p.m.	14

2. Mike administered a skills assessment to his employees. Possible scores on the skills assessment ranged from 200 to 1800. The following table shows how his employees performed on the assessment:

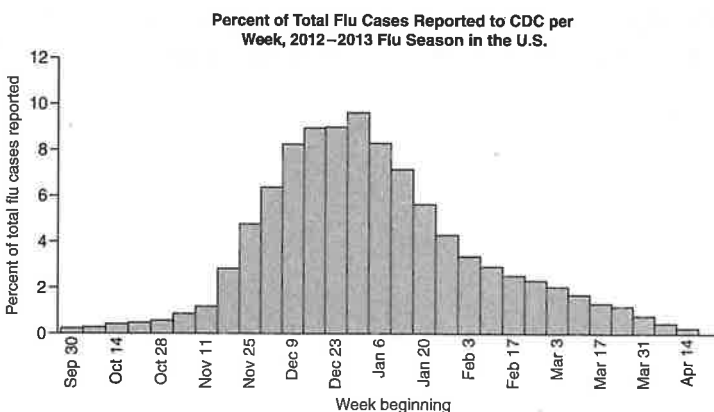
Score Range	Number of Employees
200–399	15
400–599	22
600–799	35
800–999	62
1000–1199	80
1200–1399	69
1400–1599	47
1600–1800	20



B. Choose the one best answer to each question.

Question 3 refers to the following histogram and information.

Influenza, or “flu,” season in the United States tends to last from fall through spring.



Source: Centers for Disease Control and Prevention

3. Approximately what percentage of flu cases was reported to the CDC during the 2012–2013 season during the time period of December 9 to January 5?

- A. 7%
- B. 15%
- C. 36%
- D. 50%

Answers and explanations begin on page 668.

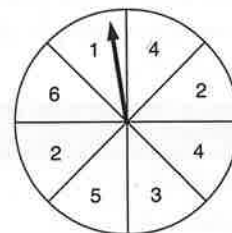
Lesson 7: Probability

Simple Probability

Probability tells whether something is likely or unlikely to happen. The probability of any event can be expressed by a number from 0 to 1. If an event has 0 probability, the event is impossible. An event with a probability of 1 is certain to happen. Most events are somewhere in between.

To find the probability of a simple random event, we must identify favorable and possible outcomes. A **favorable outcome** is the event that we are interested in. The **possible outcomes** are all the possible events that could occur. **Theoretical probability** (sometimes called **simple probability**) is the ratio of favorable outcomes to possible outcomes.

Example 1: The spinner is divided into 8 equal sections. What is the probability of spinning a 4 on the spinner?



1. There are two sections labeled 4 on the spinner, and there are 8 sections in all.
2. Use the probability ratio: $\frac{\text{favorable outcomes}}{\text{possible outcomes}} = \frac{2}{8} = \frac{1}{4}$.

The probability of spinning a 4 on the spinner is **1 out of 4, $\frac{1}{4}$, 0.25, or 25%**.

In Example 1, probability was based on what we knew could happen. Another type of probability, called **experimental probability**, is based on what actually happens during the trials of an experiment. The number of trials are the number of times you try the experiment.

Example 2: Ricardo and Scott used the same spinner to play a game. They kept track of the numbers that they got on each spin for 20 spins. The numbers are shown below.

2, 4, 4, 6, 4, 3, 4, 6, 4, 3, 1, 6, 2, 2, 5, 2, 4, 2, 1, 2

Based on their results, what is the experimental probability of spinning a 4?

1. Ricardo and Scott spun a 4 six times out of twenty.
2. Use this ratio: $\frac{\text{favorable outcomes}}{\text{number of trials in experiment}} = \frac{6}{20} = \frac{3}{10}$, **0.3, or 30%**.

Notice that experimental probability is close to, but not necessarily equal to, theoretical probability. Theoretical probability can tell you what will probably happen, but it can't predict what will actually happen.

Key Ideas

- Probability is a ratio. It can be expressed as a ratio, fraction, decimal, or percent.
- Theoretical probability is the ratio of favorable outcomes to possible outcomes.
- Experimental probability is the ratio of favorable outcomes to the number of trials in an experiment.

GED® TEST TIP

In a multiple-choice probability problem, skim the answer choices to see if they are in fraction, percent, or ratio form. Knowing the answer form will help you decide how to do your calculation.

Practice 7.1

- A. Express probability as a fraction, decimal, and percent for questions 1 through 5.
Do not use a calculator.

- A game has 50 wooden tiles. Players draw tiles to spell words. If 20 of the tiles are marked with vowels, what is the probability of drawing a vowel from the tiles?
- A spinner has five equal sections colored either red, white, or blue. After 40 spins, a player has the following results:

Color	Frequency
red	
white	
blue	

What is the experimental probability of not spinning blue on the spinner?

- There are four red, four blue, and two green marbles in a bag. If one marble is chosen at random from the bag, what is the probability that the marble will be green?
- A movie theater sells 180 adult tickets and 60 children's tickets to a movie. As part of a special promotion, one ticket will be chosen at random, and the winner will receive a prize. What is the probability that the winner will be a child?
- A spinner has six equal sections numbered from 1 to 6. What is the probability of spinning either a 5 or 6?

- B. Choose the one best answer to each question. You MAY use your calculator.

Questions 6 and 7 refer to the following information.

A deck of 12 cards is marked with the following symbols.



$$P = \frac{6}{12} = \frac{1}{2} = 50\%$$

- ✓ 6. If a card is chosen at random, what is the probability of selecting a diamond (♦)?
- A. 6%
B. 12%
C. 50%
D. 60%
7. If a card is chosen at random, what is the probability of selecting something other than a club (♣)?

A. $\frac{3}{4}$
B. $\frac{2}{3}$
C. $\frac{1}{3}$
D. $\frac{9}{100}$

$$P = 1 - \frac{3}{12} = 1 - \frac{1}{4} = \frac{3}{4} = 75\%$$

Questions 8 and 9 refer to the following information.

Erin flipped a coin 40 times and made this table to show how many outcomes were "heads" and how many were "tails."

heads	
tails	

- Based on Erin's data, what is the experimental probability of getting tails on a coin flip?
A. 3 out of 5
B. 3 out of 4
C. 2 out of 3
D. 2 out of 5
- Based on Erin's data, what is the experimental probability of getting heads on a coin flip?
A. 3 out of 5
B. 3 out of 4
C. 2 out of 3
D. 1 out of 2

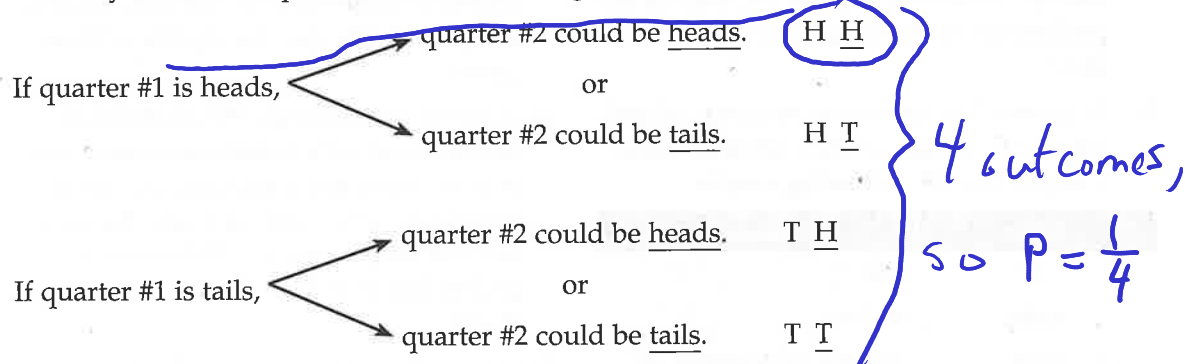
Answers and explanations begin on page 669.

Dependent and Independent Probability

You know how to find the probability of a single event. You can use this knowledge to find the probability of two or more events.

Example 3: Brad tosses two quarters into the air. What is the probability that both will land so that the heads' sides are showing?

One way to solve the problem is to list or diagram all the possible outcomes.



There are four possible outcomes, and only one is favorable (HH). Therefore, the probability of having both land with the heads side up is $\frac{1}{4}$, or 25%.

You can also use multiplication to find the probability.

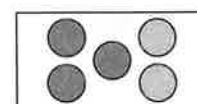
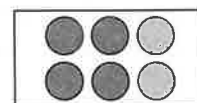
- Find the probability of the individual events. The probability that one coin will be heads is $\frac{1}{2}$, and the probability that the other will be heads is $\frac{1}{2}$.
- Multiply to find the probability of both events: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.

The two coin tosses in Example 3 are **independent events**. When events are independent, one does not affect the probability of another. In Example 4 below, the events are **dependent**. Once the first event takes place, the probability of the second event is changed.

Example 4: A box contains four blue marbles and two red marbles. If you select two marbles, what is the probability that both will be blue?

(Hint: Even though the marbles in the box are taken out at the same time, think of one as the first marble and the other as the second marble.)

- There are six marbles in the box, and four are blue. The probability that the first marble will be blue is $\frac{4}{6}$, which reduces to $\frac{2}{3}$.
- Assume the first marble selected is blue. Now there are only five marbles in the box, and three are blue. The probability that the second marble will be blue is $\frac{3}{5}$.
- Multiply to find the probability of the two events: $\frac{2}{3} \times \frac{3}{5} = \frac{6}{15}$, or $\frac{2}{5}$.



The probability that both marbles will be blue is **2 out of 5**.

NOTE: The events in Example 4 would not be dependent if the first marble were replaced before the second marble was selected. Always think carefully about the situation to decide whether two events are dependent or independent.

Practice 7.2

A. Solve as directed. Express answers as fractions.

- Kim rolls two standard six-sided dice. What is the chance that both will be 4s?
- Ten cards are numbered from 1 to 10. Toni draws out a card, replaces it, and then draws another card. What is the probability that both cards will be numbers greater than 5?
- A spinner has four equal sections. Two sections are red, one is green, and one is blue. If the spinner is spun three times, what is the probability that all three spins will be red?

- Twenty marbles are placed in a bag. Ten are red, and ten are black. One marble is drawn from the bag and set aside. Another marble is drawn from the bag. What is the chance that both marbles will be red?
- Allison tosses a coin four times. What is the chance that the coin will be heads all four times?
- If you roll two standard dice, what is the probability that both will be an odd number?

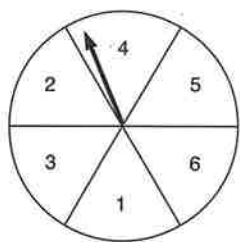
$$P = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

$$P = \frac{10}{20} \times \frac{9}{19} = \frac{1}{2} \times \frac{9}{19} = \frac{9}{38}$$

B. Choose the one best answer to each question. You MAY use your calculator.

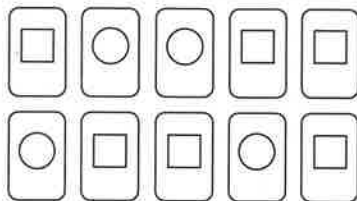
Questions 7 and 8 refer to the following information.

In a game a player rolls a die, numbered from 1 to 6, and spins a spinner. The spinner is shown below.



- What is the probability of rolling a 5 and then spinning an even number?
 - $\frac{1}{12}$
 - $\frac{1}{6}$
 - $\frac{2}{3}$
 - $\frac{5}{6}$
- What is the chance that a player will get a 3 on both the die and the spinner?
 - $\frac{5}{6}$
 - $\frac{2}{3}$
 - $\frac{1}{6}$
 - $\frac{1}{36}$

Daniel uses the ten cards below in a magic trick.



- Daniel shuffles the cards and asks an audience member to choose and hold two cards. If the cards are chosen randomly, what is the chance that both will be marked with a square?
 - 8 out of 14
 - 3 out of 5
 - 1 out of 3
 - 1 out of 5
- There are 15 colored chips in a bag. Eight are green, and seven are white. Five white chips are removed. What is the probability that the next chip selected will be green?
 - 100%
 - 80%
 - 75%
 - 25%

Answers and explanations begin on page 669.

dividing fractions:

$$\frac{3}{4} \div \frac{2}{3} = \frac{3}{4} \times \frac{3}{2}$$
$$= \frac{9}{8}$$

Lesson 8: Combinations

Combinations with One Type of Item

Key Ideas

- Combinations are used to select several things out of a larger group when the order doesn't matter.
- If you are asked to find how many ways to combine items from one group, use an organized list or table.
- If you are asked to find how many ways to combine one item from each of a number of different types of items, use a tree diagram or the fundamental counting principle.

ON THE GED® TEST

While you are taking the test on computer, you can use the provided marker and wipe-off board to make a tree diagram or other diagrams to help you visualize problems.

Sometimes the GED® *Mathematical Reasoning Test* will ask you how many ways you can combine a set of items. Sometimes you may have only one type of item to combine, and sometimes you may have more than one type to combine. This difference will determine your problem-solving strategy.

Consider an example with *only one type of item*:

Example 1: Pablo is going shopping at a fruit stand that sells apples, bananas, grapes, and pears. Pablo will buy two different kinds of fruit. How many combinations of two kinds of fruit could Pablo buy?

The question is asking you to list possible groups of two out of the overall set of four. Notice that order doesn't matter—that is, *apples and bananas* is no different from *bananas and apples*. To solve this problem, you can make a list. Start with apples and list all of the **combinations** that include apples: AB, AG, AP. Then make a list of groups that start with bananas and include the remaining fruits: BG, BP. Don't include BA because that's the same as AB. Then make a list of groups that start with grapes and include only the remaining fruits: GP. There are no remaining groups that start with P. You may find it easier to make this list in columns, as follows:

AB
AG BG
AP BP GP

$${}^nC_r = \frac{n!}{(n-r)!r!}$$

There are 6 possible combinations.

You can also draw a quick table like the following. Include a column for each of the fruits, and let each row represent a possible combination. Place two x's in each row to represent a combination of two fruits.

Apples	Bananas	Grapes	Pears
x	x		
x		x	
x			x
	x	x	
	x		x
		x	x

Notice that you could not draw any more rows with two x's without duplicating some of the existing rows. Count the rows: there are 6 possible combinations of two fruits.

Practice 8.1

A. Use counting, an organized list, or a table to solve the following problems.

1. As a supervisor, Rob is choosing four of his employees to work on a special project. The employees are Angela, Barbara, Colin, David, and Elizabeth. Of those five employees, how many teams of four are possible?
2. Jessica wants to take three books on vacation with her. She has five books to choose from. How many possible combinations of three books could she take on vacation?
3. Grant is cooking a homemade pizza, and he has the following toppings available: anchovies, ham, mushrooms, and sausage. He will choose three of those toppings. How many ways could Grant combine those toppings?
4. Five friends want to play a video game designed for two players. How many different combinations of two friends are possible, if the order of the friends does not matter?

B. Choose the one best answer to each question about combinations.

5. Celia is going to plant a small flower bed with four flowers. She can choose from begonias, fuchsias, hellebore, daisies, and salvia. How many combinations of four flowers are possible?
A. 5
B. 20
C. 24
D. 120
6. Regan owns a small business, and she wants to send three people from her business to an upcoming trade show. Since Regan plans to attend, she will only choose two of her four employees to also attend. How many combinations of employees could Regan choose to send to the trade show?
A. 4
B. 6
C. 10
D. 24
7. A book club, which has two women and three men, can send three of its members to a book signing. How many possible combinations of members could attend the book signing?
A. 1
B. 5
C. 10
D. 120
8. Given six people, how many combinations of three are possible if order does not matter?
A. 20
B. 120
C. 180
D. 720
9. The Sarkesian brothers own a hardware store. There are five items the owners might display in the window. Joe Sarkesian thinks that the window should have two items, but his brother Rick thinks the window should have three items. Counting both Joe's ideas and Rick's ideas for the window, how many possible combinations of items might go into the window display?
A. 10
B. 20
C. 30
D. 120

Answers and explanations begin on page 669.

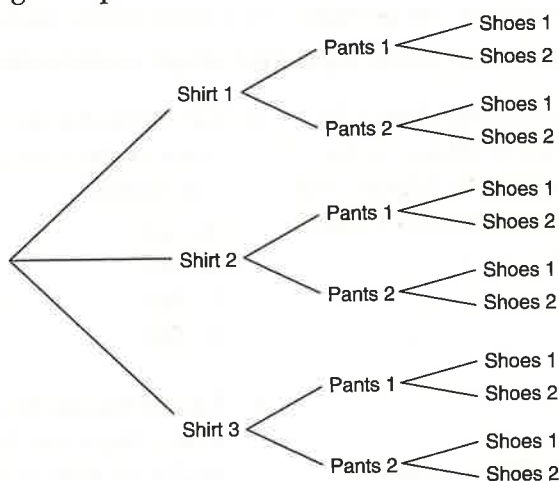
$$\begin{aligned} \rightarrow {}^5C_4 &= \frac{5!}{(5-4)!4!} \\ &= \frac{5 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times 1}{1 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times 1} \\ &= 5 \end{aligned}$$

Combinations with More Than One Type of Item

Sometimes you may be asked to count possible combinations with more than one type of item. Consider the following example:

Example 2: Sarah is deciding what to wear. She has three shirts, two pairs of pants, and two pairs of shoes. How many possible outfits does she have, if an outfit is one shirt, one pair of pants, and one pair of shoes?

This question differs from the question about Pablo on page 300. In that question, Pablo was choosing from one overall group of fruits. Here, Sarah is choosing one out of each of three groups—shirts, pants, and shoes. If you are given more than one type of item and must choose one of each type, you can use a tree diagram to figure out all the possible combinations. Study the following example:



Each “branch” of the tree represents a possible combination. The number of branches on the right-hand side represents the possible number of combinations; in this case, there are 12 possible outfits.

However, you can solve the same type of problem (where you have more than one type of item and are choosing one of each type) using the **fundamental counting principle**, which works like this:

Start with shirts: for each shirt, Sarah has two pants options. That’s $3 \times 2 = 6$. Additionally, for each of those six options, she has two options for shoes. That’s $3 \times 2 \times 2 = 12$.

Use the fundamental counting principle to solve the following problem:

Example 3: Gordon is at a restaurant that serves a three-course meal: an appetizer, an entrée, and a dessert. There are three appetizers, six entrees, and four desserts to choose from. Gordon will order one of each. How many different meals could he order?

Multiply three appetizers by six entrees by four desserts: $3 \times 6 \times 4 = 72$. There are 72 possible ways Gordon could order his meal.

Practice 8.2

- A. Use a tree diagram or the fundamental counting principle to solve the following problems.
1. Frank is making a sandwich. His sandwich will have one type of bread, one type of meat, and one condiment. He has four types of bread: white, wheat, rye, and a Kaiser roll. Meats he can choose from are chicken, turkey, and roast beef. For condiments, Frank can choose mayonnaise, mustard, vinaigrette, or horseradish. How many ways can Frank make his sandwich?
 2. Van Ahn attended a trivia contest and won three rounds. For the first round she won, she could choose one of three restaurant gift certificates. The prize for another round was one of two T-shirts. Winning the third round allowed her to choose one of four souvenir hats. How many possible combinations of prizes were available to Van Ahn?
 3. Henry is wrapping a gift for his daughter. He will use one type of wrapping paper, one type of ribbon, and one bow. He has three different types of wrapping paper, three different types of ribbon, and three different bows. How many ways can Henry wrap the gift?
 4. Julio is looking forward to a three-day weekend. On Friday, he will either go to the beach or attend a street festival. On Saturday, he plans to visit a museum, but he is trying to decide among three different museums. On Sunday, he will try one of four new restaurants. How many ways could Julio combine these weekend activities?
- B. Choose the one best answer to each question about combinations. Determine whether you are being asked to find possible combinations from one type of item or from multiple types of items. You MAY use your calculator.
5. The Nu Mu Beta fraternity is deciding what secret passphrase it should require its members to say before being admitted to meetings. The first and third words of the passphrase will be the names of Greek letters. There are 24 letters in the Greek alphabet. The second word will be a number from one to nine. How many possible passphrases are there?
A. 24
B. 57
C. 576
D. 5184
 6. A video game designer is creating a new superhero, who will have three of five potential superpowers. How many combinations of superpowers could the new superhero have?
A. 10
B. 15
C. 100
D. 125
 7. A chef will make a soup with five ingredients: one of four meats, one of four vegetables, one of four kinds of noodles, one of four kinds of broth, and one of four spices. How many possible combinations of ingredients could the chef put into the soup?
A. 20
B. 256
C. 1024
D. 3125
 8. A doctor is deciding how to treat a given disease. The doctor will prescribe one medication, one dietary change, and one type of vitamin supplement. There are five medications, five dietary changes, and five types of vitamins the doctor might prescribe. How many combinations are possible?
A. 125
B. 625
C. 1250
D. 3125

Answers and explanations begin on page 669.

Lesson 9: Permutations

Key Ideas

- Permutations are used to find the number of possible outcomes when the order does matter.
- Use the total number of people or items and the number of possible outcomes to find the number of permutations that are possible.

On the *Mathematical Reasoning Test*, you may be asked how many possible permutations, or sequences, are possible given a group of items. Consider the following example:

Example 1: Eliza is planning her day off. She wants to visit the art museum, try the new coffee shop, call her mom, and take a walk, but not necessarily in that order. How many possible sequences of those four activities are there?

This question asks you for all possible permutations, or ways to sequence the items. Thus, *coffee-art-mom-walk* and *art-coffee-mom-walk* are two different possible outcomes. Consider how this problem differs from the combinations problems on pages 300–303. In those cases, order did not matter; in a permutations problem, order does matter.

To find how many ways you can sequence all the items in a list, multiply the possibilities like this. Start with how many things could go first: here, there are four possibilities for Eliza's first activity. Once she has done that first activity, three possibilities remain. After she has done a first and second activity, two possibilities remain, and then only one. So you simply multiply: $4 \times 3 \times 2 \times 1$. There are **24** possible permutations.

Some permutations questions may ask you to determine the possible orderings for only some of the items in a list. Consider this example:

Example 2: Ten runners are competing in a race. There are prizes for first, second, and third place. How many possible sequences of the top three prize winners are there?

Notice that you are not simply figuring out how many groups of three can be made out of the ten. Rather, you are figuring out how many sequences of three can be made out of the ten. That can sometimes be represented on the GED® Test with the notation $P(10, 3)$.

Now, multiplying $10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$ would give you all the possible permutations of all ten runners, so that approach will not work here. Instead, start with how many people could win first place: here, ten. Once someone has won first, there are nine possibilities for second place. Once someone has won first and someone else has won second, there are eight possibilities for third place. And then you stop multiplying: $10 \times 9 \times 8 = 720$. There are **720** possible ways that the runners could be arranged in the top three prize-winning slots. (Note that simply multiplying 10 by 3 will not work.)

ON THE GED® TEST

When you are reading a question that suggests that you may want to find a combination or a permutation, ask yourself, "Am I finding a number of groups (combinations), or am I finding the number of possible sequences (permutations)?"

Practice 9

A. Read the problems and decide whether you are being asked to find combinations or permutations. Then solve.

1. Noemi is trying to remember the password for her email. She knows that it has the following characters in it: M, Q, \$, L, 7. But she can't remember the order they go in. How many possible sequences of those five characters are there?
2. Bilal is going to the museum. She wants to see the contemporary art exhibit, the Impressionists, the Greek vases, and a special Picasso exhibit but not necessarily in that order. How many sequences of those exhibits are possible?
3. Five students in a class have volunteered for a special project. Only three can actually help with the project. How many groupings of three out of the five are possible?
4. Tyrell is the curator at a gallery and is deciding how to arrange six paintings that will be displayed in a line along one wall. All six paintings will be included. How many possible sequences could Tyrell choose?

B. Choose the one best answer to each question about combinations or permutations. You MAY use your calculator.

5. Soraya has been given six tasks to do at work, but she has time to complete only four of them. She must decide in what order to do the tasks. How many possible orderings of four tasks are available to Soraya?

A. 24
B. 36
C. 360
D. 720

6. In a certain public garden, the gardener wanted to show the different visual effects that arranging flowers in different sequences can have. So he chose three kinds of flowers and planted flower beds showing each of the possible sequences of the three kinds. How many such flower beds did the gardener plant?

A. 3
B. 6
C. 9
D. 12

7. Ten people hope to become extras in a movie. The movie's casting director will choose four people to fill the following specific roles in the movie: Bystander #1, Bystander #2, Bystander #3, and Bystander #4. How many ways could those four roles be filled?

A. 24
B. 40
C. 540
D. 5040

Questions 8 and 9 refer to the following information:

Clark and his daughter are at the amusement park, and Clark is offering his daughter a choice. They don't have time to ride all six rides at the amusement park, but they can ride three of them.

8. How many combinations of three rides could Clark's daughter choose?

A. 18
B. 20
C. 120
D. 720

9. Clark's daughter can also choose the order in which she wants to enjoy the rides. How many possible orderings of three out of the six rides are possible?

A. 18
B. 20
C. 120
D. 720

Answers and explanations begin on page 670.

Lesson 10: Compare Proportions in Different Formats

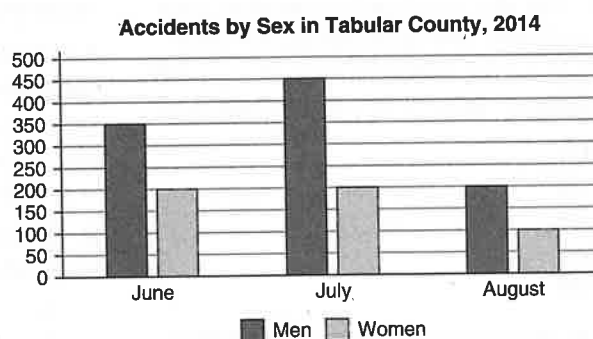
Key Ideas

- A proportion can be represented in multiple ways, but its magnitude does not change.
- It is easiest to compare the magnitude of two values if they are represented in the same way.

The relative magnitude of two quantities can be represented in multiple ways: as simple numbers, on a graph or chart, or as dimensions on a blueprint or diagram. It is harder to compare two quantities when they are presented differently—for example, when one is represented on a bar graph and the other is represented in a table of values.

To directly compare quantities represented differently, translate them into the same format. Typically, using a numerical representation is most efficient for both translation and comparison purposes.

Example 1: The bar graph shows the number of accidents for men and women aged 18–35 in Tabular County for the summer of 2014. The following year, the county released a report that the monthly rate of accidents for men in 2015 had been 1.6, 2.3, and 1.8 times that of women for June, July, and August, respectively. Of the six months reported, when was an accident victim most likely to be male?

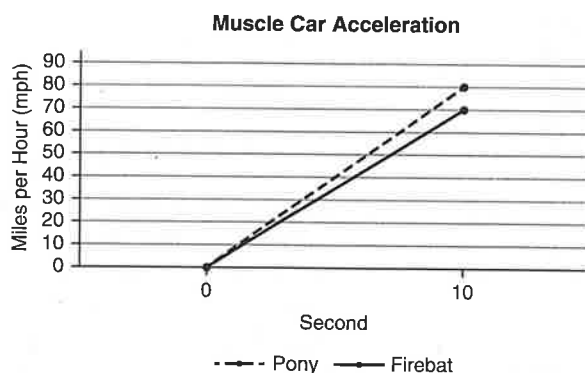


GED® TEST TIP

When a question asks you to compare two proportions presented in different formats, convert one or both proportions so they are in the same format. Usually, proportions in a numerical format are easiest to compare.

1. Determine the proportion of 2014 male accidents to female accidents from the bar graph. For June, the ratio is $\frac{350}{200} = 1.75$, for July $\frac{450}{200} = 2.25$, and for August $\frac{200}{100} = 2$.
2. Compare the calculated and given values to determine the greatest proportion of male to female accidents. The highest value for this ratio is in July 2015, when men were 2.3 times as likely to have an accident as women.

Example 2: The following velocity-time graph shows the rate of acceleration of two classic muscle cars. A newer model, the Hermes, is able to accelerate from 0 to 60 miles per hour (mph) in 8 seconds (s).



How does the Hermes compare to the cars on the graph in terms of acceleration?

1. Use the formula for slope of a line, $(m) = \frac{\text{rise}}{\text{run}}$, to find the rate of acceleration for the cars on the graph. The Pony's acceleration is $\frac{80-0}{10-0} = 8\text{mph/s}$, and the Firebat's is $\frac{70-0}{10-0} = 7\text{mph/s}$.
2. Write the rate of acceleration of the Hermes the same way, as $\frac{60}{8} = 7.5\text{mph/s}$.
3. Compare the values to determine the relative speeds of the cars. The Hermes is faster than the Firebat but still slower than the Pony.

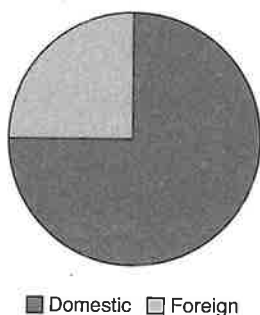
Practice 10

Compare the proportional data in the following scenarios.

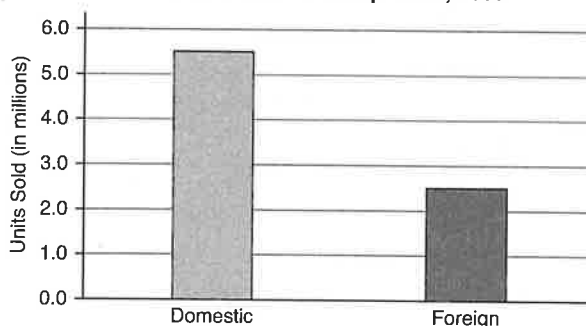
Question 1 is based on the following graphs.

1. The two graphs show domestic and foreign sales for a brand of US potato chips. For which year were the foreign sales higher as a percentage of total sales, and what was the difference in that percentage between the years?

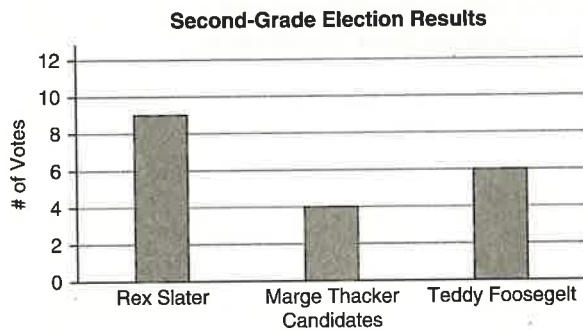
Brand X Potato Chip Sales, 1975



Brand X Potato Chip Sales, 2005

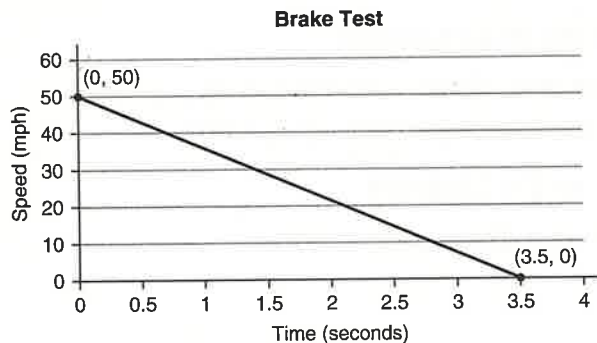


Question 2 is based on the following graph.



2. A second-grade political candidate, Rex Slater, predicted he would win 55 percent of the class vote. The results of the election are shown in the bar graph. Was he correct?

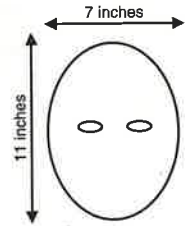
Question 3 is based on the following graph.



3. Engineers at an automotive parts company are testing brakes to find out whether their brakes beat those of their competitors, which can decelerate the same car at a rate of 15 miles per hour (mph) per second (s). Which statement correctly describes the results of their latest test, shown on the graph?
- The brakes are better than the competitors' brakes by more than 1 mph/s.
 - The brakes are better than the competitors' brakes by less than 1 mph/s.
 - The brakes are worse than the competitors' brakes by less than 1 mph/s.
 - The brakes are worse than the competitors' brakes by more than 1 mph/s.

Question 4 is based on the following chart and figure.

Model	Ratio
A	1.50
B	1.65
C	1.70
D	1.56
E	1.53



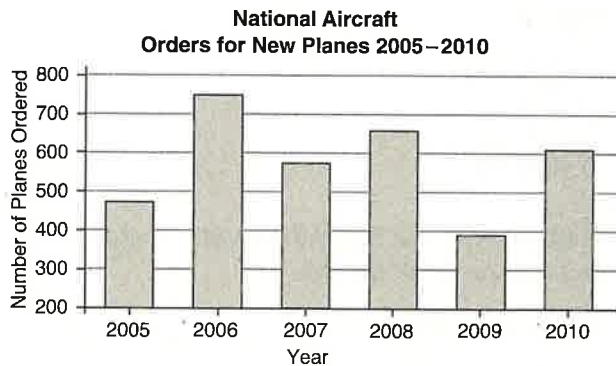
4. A group of psychologists are testing an ancient Greek theory about beauty. According to this theory, the ratio of the height of the face to the width of the face should be as close to 1.6 as possible. The chart above shows the results for some pictures of models in magazines. Given the face shown, determine how beautiful the person is compared to the models.
- The person is more beautiful than any of the models.
 - The person is more beautiful than four of the models.
 - The person is more beautiful than three of the models.
 - The person is more beautiful than two of the models.

Answers and explanations begin on page 670.

Data, Statistics, and Probability Practice Questions

Directions: You MAY use your calculator.

Questions 1 and 2 refer to the following graph.



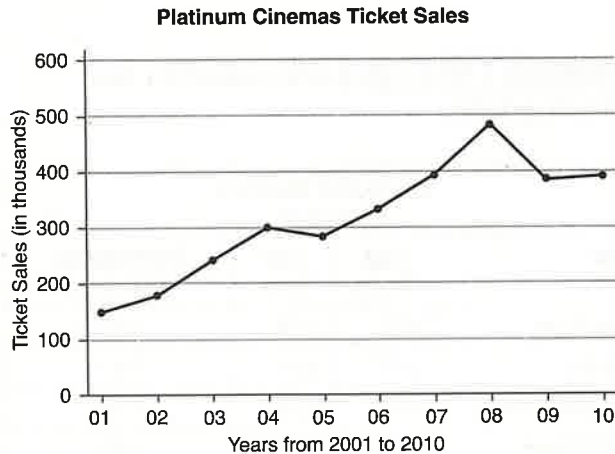
- The mean number of aircraft orders for the six years shown on the graph is 573 planes. In which year was the number of orders closest to the mean?
A. 2006
B. 2007
C. 2008
D. 2009
- By about what percent did orders at National Aircraft decrease from 2008 to 2009?
A. 30%
B. 40%
C. 68%
D. 75%
- At a convention, Jim and his three friends each bought three raffle tickets. At the time of the drawing, 400 tickets had been sold. What is the probability that either Jim or one of his friends will win?
A. $\frac{3}{100}$
B. $\frac{1}{25}$
C. $\frac{3}{50}$
D. $\frac{9}{100}$

Questions 4 through 6 are based on the following table.

Southland Weather March 9			
Area	High Temp.	Low Temp.	Precipitation (in inches)
Downtown	65°F	53°F	0.45
Airport	62°F	50°F	0.63
Woodland Hills	68°F	50°F	1.34
East Village	56°F	48°F	3.53
Ventura	62°F	49°F	2.57
Highland Park	64°F	55°F	0.84

- Based on the data in the table, what was the median low temperature for March 9?
A. 62.8°
B. 51.5°
C. 50.8°
D. 50°
- What was the mean amount of precipitation (in inches) on March 9 for the areas listed in the table?
A. 0.65
B. 1.09
C. 1.56
D. 1.99
- For which area on the table was there the greatest range, or difference, between the high and low temperatures?
A. Ventura
B. East Village
C. Woodland Hills
D. Downtown

Questions 7 and 8 refer to the graph.



7. Platinum Cinemas opened its first theaters in 2001. What was the first year after 2001 in which the company experienced a decline in sales?
- A. 2004
B. 2005
C. 2008
D. 2009
8. Which year had the sharpest increase in ticket sales over the previous year?
- A. 2002
B. 2006
C. 2008
D. 2009

9. At Nelson Stationers, the first 25 customers who visited the store on Monday morning received their choice of a gift. The table below shows how many customers chose each gift.

Pen and Pencil Set	
Calculator	
Mouse Pad	

What percent of the customers chose a mouse pad?

- A. 17%
B. 25%
C. 32%
D. $33\frac{1}{3}\%$

10. A standard deck of playing cards has 52 cards, with 13 cards each of hearts, diamonds, clubs, and spades. If a card is drawn randomly from the deck, what is the probability that it will be either hearts or diamonds?
- A. 1 in 2
B. 1 in 4
C. 1 in 8
D. 1 in 16

11. Nita worked the following overtime hours over a six-week period.

Week 1: 5 hours

Week 2: $3\frac{1}{2}$ hours

Week 3: 4 hours

Week 4: 0 hours

Week 5: $1\frac{1}{2}$ hours

Week 6: 7 hours

What is the mean number of overtime hours Nita worked each week?

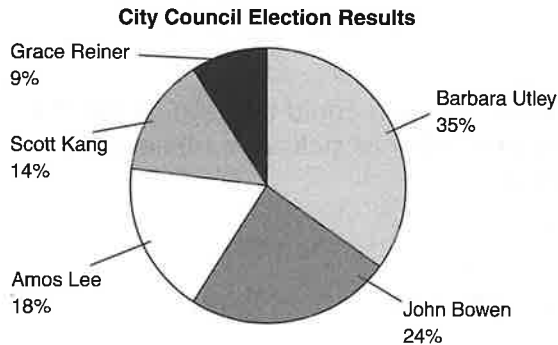
Write your answer on the line below.

12. A spinner has five equal sections, and they are numbered from 1 to 5. What is the probability of spinning a number greater than 3? (Express the answer as a fraction.)

Write your answer on the line below.

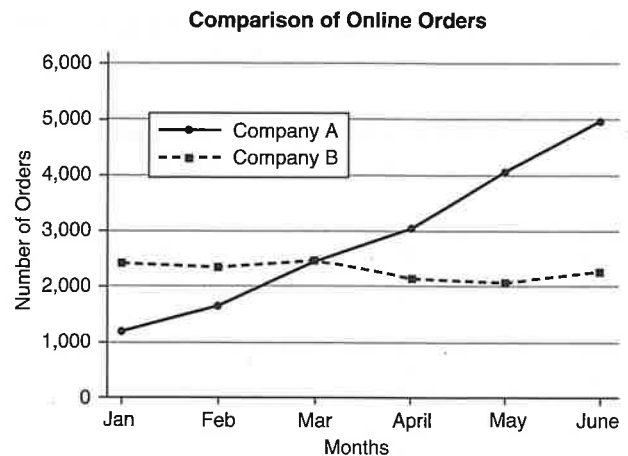
Questions 13 through 15 refer to the following graph.

In a recent election, five candidates ran for the city council seat from District 11. The results of the race are shown in the graph below.



13. The three city council candidates who received the fewest votes received what percent of the total vote?
 - A. 23%
 - B. 41%
 - C. 56%
 - D. 77%
14. Which two candidates combined received about $\frac{3}{5}$ of the votes cast?
 - A. Reiner and Utley
 - B. Kang and Utley
 - C. Bowen and Utley
 - D. Lee and Bowen
15. If 5100 votes were cast in the election, which of the following expressions could be used to find out how many votes Grace Reiner received?
 - A. 5100×0.9
 - B. $\frac{5100}{0.9}$
 - C. $\frac{5100}{0.09}$
 - D. 5100×0.09

Questions 16 through 18 refer to the following graph.



16. In which month did Company A and Company B receive about the same number of online orders?
 - A. February
 - B. March
 - C. April
 - D. May
17. Based on the trends shown by the data, what would be the best prediction for the number of online orders in July for Company A?
 - A. 7200
 - B. 5900
 - C. 5200
 - D. 4900
18. About 18 percent of the online orders in April at Company A are returned for credit or exchange. About how many orders at Company A were returned for credit or exchange in the month of April?
 - A. 180
 - B. 360
 - C. 540
 - D. 600

Questions 19 and 20 are based on the following information.

A basketball player's statistics for an eight-game series are shown in the table below.

Game	Shots Attempted	Shots Made
1	25	10
2	23	12
3	26	10
4	24	13
5	29	15
6	18	7
7	24	12
8	27	10

19. What was the median number of shots attempted by this player in the series?

- A. 24
- B. 24.5
- C. 25
- D. 26.5

20. What is the mode of the shots made by this player during the series?

- A. 12
- B. 11
- C. 10 and 12
- D. 10

21. A bag contains 24 marbles. Eight are red, six are blue, and ten are white. A marble is drawn from the bag and replaced. A second marble is chosen at random from the bag. What is the probability that the first marble is red and the second is white?

- A. $\frac{2}{5}$
- B. $\frac{5}{36}$
- C. $\frac{5}{48}$
- D. $\frac{1}{48}$

22. A company has 36 employees. For three months, the owner has kept track of the number of sick days used by her employees per month.

Month	Sick Days
Sept.	34
Oct.	31
Nov.	42

Which expression could be used to find the average number of sick days taken each month?

- A. $\frac{34 + 31 + 42}{36}$
- B. $\frac{34 + 31 + 42}{3}$
- C. $\frac{34 + 31 + 42 + 36}{4}$
- D. $(34 + 31 + 42) \times 3$

For questions 23 through 27, write your answers on the lines provided.

23. Andy rolls two standard six-sided dice. What is the probability of rolling two 1s? Express your answer as a fraction.

24. Kate drove 158 miles on Monday, 276 miles on Tuesday, 54 miles on Wednesday, 305 miles on Thursday, and 210 miles on Friday. In miles, what was the median distance she drove for the five days?

25. Angelique, a teacher, is making a reading list for her students. She wants to include one historical novel, one work of science fiction, one book of poetry, and one graphic novel. She has five of each type of book to choose from. How many possible combinations of books could she include in her reading list?

26. There are four children in Mr. Martin's class. Three of them will be selected to be hall monitors. How many combinations of hall monitors are possible?

27. A charity's board has ten members. One of the members will be elected president, then another will be elected treasurer, and then a third will be elected secretary. If no member can hold more than one position, how many ways can the members fill those positions?

Questions 28 and 29 are based on the following information.

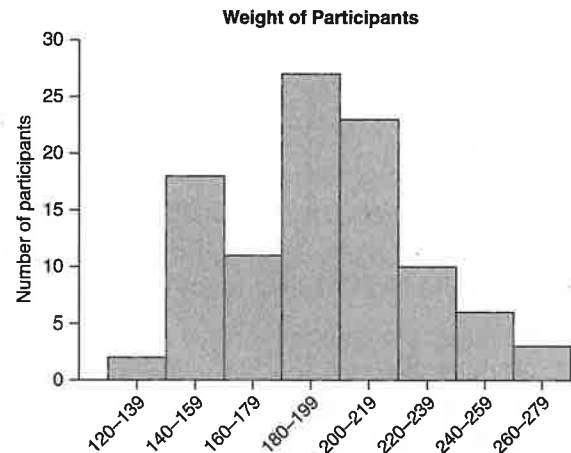
The frequency table below displays the reasons why patients visited a walk-in clinic during a certain week.

Reason	Number of Patients
Colds and Flu	
Cuts and Scrapes	
Sprained Muscles	
Tetanus Shots	
Severe Headaches	

28. Write your answer on the line below. What percent of the patients represented in the table visited the walk-in clinic because of sprained muscles?
29. In the week represented in the table, what was the ratio of the number of patients with colds or flu to the number of patients with severe headaches?
- A. 4:15
B. 4:3
C. 3:4
D. 1:5

Questions 30 and 31 are based on the following information and graph.

A hundred people are participating in a medical study. As a first step, the researchers collected information about participants' weight. The graph below represents how many people fell into each weight category.



30. Which weight category displays the highest frequency?
- A. 120–139 lbs
B. 180–199 lbs
C. 200–219 lbs
D. 260–279 lbs
31. Consider only the 100 individuals represented in the graph. Which one of the following groups contains the most people?
- A. Participants weighing 200 lbs or more
B. Participants weighing less than 200 lbs
C. Participants weighing 180–199 lbs
D. Participants weighing less than 180 lbs

Questions 32 through 34 refer to the following information and table.

The owner of a taxi company measures the gas mileage of each of the cars in her fleet. The frequency table below shows her results.

Average Miles per Gallon	Number of Cars
15	11
20	4411
25	441
30	4411
35	1

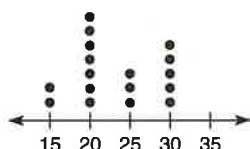
32. Which of the values (average miles per gallon) displays the highest frequency?

- A. 15 miles per gallon
- B. 20 miles per gallon
- C. 30 miles per gallon
- D. 35 miles per gallon

33. What is the ratio of the number of taxis that get 35 miles per gallon on average to the number of taxis that get 15 miles per gallon on average?

- A. $\frac{1}{2}$
- B. $\frac{3}{2}$
- C. $\frac{4}{5}$
- D. $\frac{1}{10}$

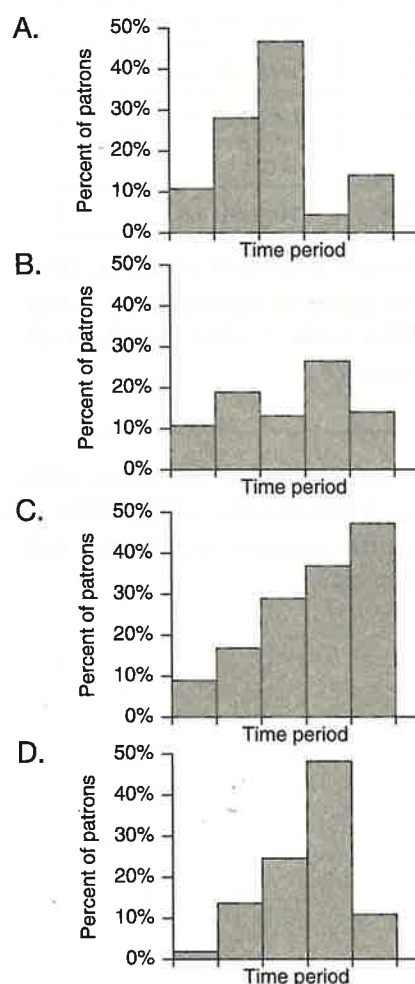
34. The line plot below is based on the information in the table. Place additional points on the line to accurately represent the data.



35. Ahmed works at the ticket window of a theater, and one night he decides to track when the patrons arrive for a 7:30 p.m. show. The table below represents his data.

Time period	Percent of Patrons who Arrive
5:30 to 5:59 p.m.	2%
6:00 to 6:29 p.m.	14%
6:30 to 6:59 p.m.	25%
7:00 to 7:29 p.m.	48%
7:30 to 8:00 p.m.	11%

Which one of the histograms below accurately represents Ahmed's data?



Answers and explanations begin on page 670.

Lesson starts on the next page ► ► ►

Algebra Basics, Expressions, and Polynomials

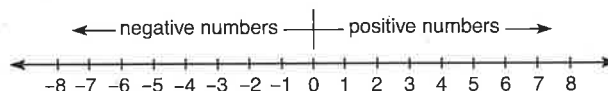
Lesson 1: The Number Line and Signed Numbers

Key Ideas

- To add numbers with like signs, add and keep the same sign.
- To add numbers with unlike signs, find the difference and use the sign from the larger number.
- To subtract signed numbers, change the operation to addition and change the sign of the number you are subtracting.

Understanding Signed Numbers

Signed numbers include zero, all positive numbers, and all negative numbers. Zero is neither positive nor negative. On a number line, the positive numbers are shown to the right of zero, and the negative numbers are shown to the left.



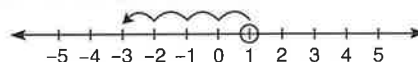
A positive number may be written with a plus (+) symbol. If a number has no symbol at all, we assume that it is positive. A negative number *must* be preceded by a minus (−) symbol.

A signed number provides two important facts. The sign tells the direction from zero, and the number tells you the distance from zero. For example, -5 lies five spaces to the left of zero, and $+4$ lies four spaces to the right of zero.

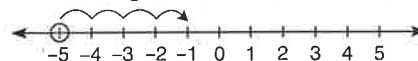
Adding and Subtracting Signed Numbers

You can use a number line to model the addition of signed numbers.

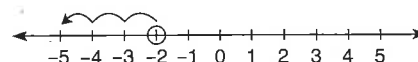
Examples: $1 + (-4) = -3$ Begin at $+1$; move 4 in a negative direction (left).



$-5 + 4 = -1$ Begin at -5 ; move 4 in a positive direction (right).



$-2 + (-3) = -5$ Begin at -2 ; move 3 in a negative direction (left).



GED® TEST TIP

You can use the “change sign” key (−) at the bottom of the online calculator. This changes a positive to a negative number, and vice versa.

To add without a number line, follow these steps:

- If numbers have like signs, add the numbers and keep the same sign.
- If the numbers have unlike signs, find the difference between the two numbers and use the sign of the larger number.

Example 1: Add $15 + (-25)$.

1. Since the numbers have unlike signs, subtract: $25 - 15 = 10$.
2. Use the sign from the larger number: $15 + (-25) = -10$.

Subtraction is the opposite of addition. To rewrite a subtraction problem as an addition problem, change the operation symbol to addition and change the sign on the number you are subtracting. Then apply the rules for adding signed numbers.

Example 2: Subtract $3 - 8$.

1. Change the operation symbol and the sign of the number you are subtracting:

$$3 - 8 \text{ becomes } 3 + (-8).$$

2. Add: $3 + (-8) = -5$.

You can use the same rules to combine several signed numbers.

Example 3: $(-5) + 6 - 4 - (-2) = ?$

1. Rewrite each subtraction as addition: $(-5) + 6 + (-4) + 2$.

2. Add the positive terms: $6 + 2 = 8$. Add the negative terms: $-5 + (-4) = -9$.

3. Combine the results: $8 + (-9) = -1$.

Practice 1.1

A. Solve.

1. $8 + (-3)$

2. $50 - 5$

3. $11 - (-2)$

4. $-1 + 2$

5. $-4 - (-5)$

6. $8 - (-2)$

7. $6 - 9$

8. $2 + 11$

9. $(-7) - (-3)$

10. $(-4) + 6$

11. $-15 + (-7)$

12. $36 - 4$

13. $-60 - (-10)$

14. $-5 - 6$

15. $12 + 13$

16. $-55 + 20$

17. $7 + (-3) + (-5) - 10$

18. $66 + (-22) - 33$

19. $-14 - (-6) + 18$

20. $80 - (-15) - 20$

21. $6 - (-3) + (-5) + 8$

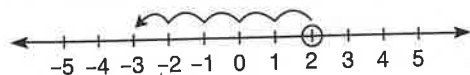
22. $-23 + (-11) - (-15) + 21$

23. $3 + 9 - 5 + 12 - 9 - 11$

24. $-7 - 20 - (-14)$

B. Choose the one best answer to each question.

Question 25 refers to the following number line.



25. The number line above shows which of the following expressions?

A. $2 + (-3)$

B. $2 + (-5)$

C. $-3 + 2$

D. $-3 - (+2)$

26. At noon, the temperature in the high desert was 92°F . A scientist observed the following temperature changes over the course of the next two hours: $+12^{\circ}\text{F}$, -5°F , $+6^{\circ}\text{F}$, -3°F , and $+13^{\circ}\text{F}$. What was the temperature, in degrees Fahrenheit at the end of the two-hour period?

A. 95°F

B. 103°F

C. 115°F

D. 131°F

Answers and explanations begin on page 671.

Multiplying and Dividing Signed Numbers

In algebra, multiplication is not shown using the times sign ($\times$) because the symbol could be easily mistaken for the variable x . Instead, multiplication is shown using a dot or by placing two numbers next to each other. To avoid confusion, one or both of the numbers may be enclosed in parentheses. For example, the expressions $-5 \cdot 6$ and $-5(6)$ and $(-5)(6)$ all mean “ -5 times 6 .”

In algebra, division can be written with the $\div$ symbol, but it is usually shown with a line that means “divided by.” The expression $\frac{20}{-5}$ means “ 20 divided by -5 ,” as does $20/-5$.

When multiplying or dividing signed numbers, use the following rules:

- If the signs are the same, the answer is positive.
- If the signs are different, the answer is negative.

Example 1: Multiply $4(-25)$.

1. Multiply the numbers only: $4 \times 25 = 100$.
2. Determine the sign. Since the signs on the numbers are different, the answer is negative:
 $4(-25) = -100$.

Example 2: Divide $\frac{-160}{-8}$.

1. Divide the numbers only: $160 \div 8 = 20$.
2. Determine the sign. Since the signs on the numbers are the same, the answer is positive:

$$\frac{-160}{-8} = 20.$$

A problem may contain more than two factors. Remember that each pair of negative factors will equal a positive product. Therefore, if there is an even number of negative terms, the product will be positive. If there is an odd number of negative terms, the product will be negative.

Example 3: $(-5)(6)(-1)(-2)(2) = ?$

1. Multiply: $5 \times 6 \times 1 \times 2 \times 2 = 120$.
2. There are three negative terms. Since 3 is an odd number, the product is negative: -120 .

When a problem contains more than one operation, follow this **order of operations**. Do multiplication and division operations first, working from left to right. Do addition and subtraction operations last, also working from left to right. If a problem contains division presented with a division bar, do any operations above and below the bar first and then divide.

Example 4: $(-4)(6) - \frac{3 + (-9)}{-2}$

1. Multiply: $-24 - \frac{3 + (-9)}{-2}$.
2. Do the operation above the fraction bar: $-24 - \frac{-6}{-2}$.
3. Divide: $-24 - (+3)$.
4. Subtract: $-24 - (+3) = -24 + (-3) = -27$.

NOTE: Notice how parentheses clarify meaning: $4 - 5$ means “four minus five,” but $4(-5)$ means “four times negative five.”

Practice 1.2

A. Solve. You MAY NOT use a calculator.

1. $(5)(4)$
2. $(7)(-3)$
3. $(-8)(6)$
4. $(-2)(-9)$
5. $(-10)(1)$
6. $9 \div 3$
7. $12 \div (-4)$
8. $-25 \div 5$
9. $(-18) \div (-9)$
10. $\frac{40}{-8}$
11. $(14)(-2)$
12. $(-75) \div 25$
13. $13 \div (-13)$
14. $(-5)(15)$
15. $\frac{18}{3}$

B. Solve. You MAY use a calculator.

16. $\frac{25(4)}{-5}$
17. $(-3)(-5)(2)(-10)$
18. $20 \div (-5) \div (-2)$
19. $\frac{6(5)}{(-3)(2)}$
20. $(-11)(2)(-5)(6)$
21. $(12)(-2) \div (-2)$
22. $(-4)(-6)(-5)$
23. $50 \div (2)(-5)$
24. $(-1)(2)(-3)(2)(-1)$
25. $\frac{(3)(-4)(2)(5)}{-6}$
26. $\frac{4(-4)}{-8(-2)}$
27. $(-5)(-2)(0)(-1)$

C. Choose the one best answer to each question.

28. Janice is creating a computer spreadsheet. A portion of her work is shown below.

	A	B	C
1	-3	4	7
2	2	-5	-8
3	-1	3	-2

Using the information from the spreadsheet, what is the value of the expression $A1 * C1 * A3 / (B3 * A3)$? (Note: In a spreadsheet, the symbol * means multiplication.)

- A. -21
- B. -7
- C. $\frac{-1}{7}$
- D. 21

29. The product of 2 and 8 is divided by -8. Which of the following expressions could be used to find the value of the statement?

- A. $\frac{2}{8}$
- B. $2(8)(-8)$
- C. $\frac{2(8)}{(-8)}$
- D. $\frac{(-8)}{2(8)}$

30. Which of the following is a true statement about the value of the expression $(52)(-103)(-45)(-8)(3)$?

- A. The result is a fraction.
- B. The result is greater than 1.
- C. The result is a negative number.
- D. The result is a positive number.

Answers and explanations begin on page 671.

Lesson 2: Powers and Roots

Key Ideas

- In the expression 2^3 , the base is 2 and the exponent is 3.
- To raise a number to a certain exponent, write the base the number of times shown by the exponent and then multiply.
- To find a square root, think, "What number times itself equals this number?"

Powers are a special way to show repeated multiplication. For example, suppose you needed to multiply $5 \times 5 \times 5 \times 5$. This series of operations can be expressed as "five to the fourth power." In other words, the number 5 appears in the multiplication problem four times.

We can write the operations algebraically using **exponents**. In the expression $5 \times 5 \times 5 \times 5$ above, the number 5 is the base. The exponent, a small number written above and to the right of the base, tells how many times the base is repeated: $5 \times 5 \times 5 \times 5 = 5^4$.

To evaluate an expression, perform the multiplication indicated by the exponent.

Example 1: Find the value of 2^5 .

Write the base the number of times indicated by the exponent and then multiply:

$$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32.$$

You may encounter some special uses of exponents on the *Mathematical Reasoning Test*. An exponent can be any positive number, or it can be 0, 1, or a negative number. Memorize the situations described below.

1. A number raised to the first power equals itself: $8^1 = 8$.
2. A number other than zero raised to the power of zero equals 1: $6^0 = 1$.
3. A number raised to a negative exponent is equal to a fraction with a numerator of 1: $4^{-2} = \frac{1}{4^2} = \frac{1}{4 \times 4} = \frac{1}{16}$.

You can use a calculator to raise numbers to any power. Use the x^2 or $\wedge$ key.

Example 2: Find the square of 24.

Press: 24 x^2 enter . The right side of the display reads 576.

The square of 24 is 576.

Example 3: Find the value of 9^4 .

Enter the base, press $\wedge$, and enter the exponent. Then press enter .

Press 9 $\wedge$ 4 enter . The right side of the display reads 6561.

Nine raised to the fourth power is 6561.

NOTE: To raise a negative number to a power using the TI-30XS MultiView™ calculator, you must enter that number in parentheses: $(-4)^2$ enter .

To square a number, multiply the number by itself. For example, $6^2 = 6 \times 6 = 36$. In the expression $6 \times 6 = 36$, the number 36 is the square, and the number 6 is the **square root** of 36.

GED® TEST TIP

Make a list of common square roots by squaring the numbers from 1 to 15. Memorize them. You will need to know common square roots to solve geometry problems about area and right triangles.

The symbol for square root is $\sqrt{\quad}$. To find a square root, think, "What number multiplied by itself is equal to the number in the bracket?"

Example 4: Find the value of $\sqrt{144}$.

You know that $12 \times 12 = 144$, so the square root of 144 is **12**.

Although $(-12) \times (-12)$ also equals 144, you will only be expected to find positive roots on the GED® Test.

You may have to approximate the value of a square root.

Example 5: What is the square root of 90?

You know that $9 \times 9 = 81$ and $10 \times 10 = 100$. Therefore, the square root of 90 is **between 9 and 10**.

You can also use your calculator to find a square root. On the TI-30SX MultiView™, you must first press the $\boxed{2\text{nd}}$ key to access the square root function. The $\sqrt{\quad}$ function is directly above the $\boxed{x^2}$ key. (Other calculators may not require the use of the $\boxed{2\text{nd}}$ key.)

Example 6: Use your calculator to find $\sqrt{90}$ to the nearest tenth.

On the TI-30SX MultiView™, press $\boxed{2\text{nd}} \boxed{x^2} 90 \boxed{\text{enter}}$. The right side of the display reads $3\sqrt{10}$.

Press the toggle key, $\boxed{\leftarrow \rightarrow}$, to change the format of the answer into a decimal.

The right side of the display now reads 9.486832981.

Rounding to the tenths place, $\sqrt{90} \approx 9.5$.

Practice 2

A. Solve each expression. You MAY NOT use a calculator.

- | | | | |
|---------------|----------------|-------------|------------------|
| 1. 3^2 | 4. 25^0 | 7. 5^3 | 10. $\sqrt{64}$ |
| 2. 4^1 | 5. $(-3)^2$ | 8. 4^{-2} | 11. 10^{-3} |
| 3. $\sqrt{9}$ | 6. $\sqrt{49}$ | 9. 2^4 | 12. $\sqrt{121}$ |

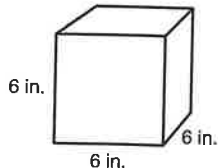
B. Solve each expression below. You MAY use your calculator. Round your answer to the nearest tenth.

- | | | | |
|------------------|-----------------|------------------|------------------|
| 13. 3^8 | 16. 20^3 | 19. $\sqrt{242}$ | 22. $\sqrt{536}$ |
| 14. $(-6)^4$ | 17. 1^{15} | 20. $(3.3)^2$ | 23. 112^0 |
| 15. $\sqrt{150}$ | 18. $(-4)^{-2}$ | 21. $\sqrt{57}$ | 24. $(-2)^8$ |

C. Choose the one best answer to each question.

25. The cube shown below measures 6 inches on each side. You can find the volume of the cube by multiplying length $\times$ width $\times$ height. Which of the following expressions represents the volume of the cube, in cubic inches?

- A. 6^1
B. 6^2
C. 6^3
D. 6^6



26. Which of the following expressions has the least value?

- A. 3^{-3}
B. 4^0
C. 4^1
D. 2^{-4}

Answers and explanations begin on page 672.

Lesson 3: Scientific Notation

Scientific notation uses the powers of ten to express very small and very large numbers. In scientific notation, a decimal number (greater than or equal to 1 and less than 10) is multiplied by a power of ten.

Look for patterns as you review these powers of ten.

$$10^1 = 10 \quad 10^2 = 100 \quad 10^3 = 1,000 \quad 10^4 = 10,000 \quad \text{and so on.}$$

$$10^{-1} = 0.1 \quad 10^{-2} = 0.01 \quad 10^{-3} = 0.001 \quad 10^{-4} = 0.0001 \quad \text{and so on.}$$

Did you find the patterns? In the row with positive exponents, the exponent is the same as the number of zeros in the number written in standard form. In the row with negative exponents, the exponent is the same as the number of decimal places in the number.

You can use these patterns to change scientific notation to standard form.

Example 1: Write 6.2×10^5 in standard form.

$$6.20000$$

Move the decimal point five places to the *right* (the same number as the exponent). Add zeros as needed.

$$620,000$$

Example 2: Write 3.82×10^{-3} in standard form.

Move the decimal point three places to the *left* (the same number as the exponent). Add zeros as needed.

$$0.00382$$

Work backward to write large and small numbers in scientific notation.

Example 3: To reach Mars, the Viking 2 spacecraft traveled 440,000,000 miles. What is the distance traveled in scientific notation?

1. Move the decimal point to the left until there is only a single digit to the left of the decimal point.

$$4.40000000$$

2. Multiply by 10 raised to a power equal to the number of places you moved the decimal point.

$$4.4 \times 10^8$$

Example 4: Scientists find that a kind of bacteria moves at a rate of 0.00016 kilometers per hour. Write the measurement in scientific notation.

1. Move the decimal place to the right until there is a single digit to the left of the decimal point.

$$0.00016$$

2. Multiply by 10 raised to a negative exponent equal to the number of places you moved the decimal point.

$$1.6 \times 10^{-4}$$

Key Ideas

- In scientific notation, a number equal to or greater than 1 and less than 10 is multiplied by a power of ten.
- Very small numbers are expressed by multiplying by a negative power of ten.
- In scientific notation, the number with the greatest power of ten has the greatest value.

GED® TEST TIP

Always try to save time. Suppose the multiple-choice answers are written in scientific notation. Instead of changing each choice to standard notation, change your answer to scientific notation.

You may be asked to compare numbers in scientific notation.

Example 5: Which is greater: 4.5×10^3 or 9.8×10^4 ?

You don't need to change the numbers to standard notation. Simply consider the powers of ten. Multiplying by 10^4 , or 10,000, must have a greater result than multiplying by 10^3 , which equals 1,000. In scientific notation, the number with the greater power of 10 has the greater value. Therefore, 9.8×10^4 is greater than 4.5×10^3 .

Practice 3

A. Write each number in scientific notation.

- | | | |
|---------------|-------------------|--------------------|
| 1. 2300 | 4. 14,320,000,000 | 7. 0.00000058 |
| 2. 0.00042 | 5. 36,000,000 | 8. 150,000,000,000 |
| 3. 12,400,000 | 6. 0.0095 | 9. 0.000000009 |

B. Convert from scientific notation to standard notation.

- | | | | |
|----------------------------|--------------------------|--------------------------|---------------------------|
| 10. 5.173×10^{-4} | 12. 4.8×10^8 | 14. 7.2×10^{-3} | 16. 8.591×10^7 |
| 11. 3.7×10^6 | 13. 1.7×10^{-5} | 15. 9.16×10^5 | 17. 9.56×10^{-6} |

C. Answer the following questions.

- | | |
|---|--|
| 18. Many domestic satellites maintain an orbit approximately 23,500 miles above Earth. What is that distance, in miles, in scientific notation? | 20. The average distance of Neptune from Earth is 2.67×10^9 miles. Write the distance, in miles, in standard notation. |
| 19. Modern technology measures very fast transactions in nanoseconds. One nanosecond equals 1.0×10^{-9} of a second. How many seconds is a nanosecond, in standard notation? | 21. Light in the vacuum of space travels at a speed of nearly 300 million meters per second. Write the speed, in meters, in scientific notation. |

D. Choose the one best answer to each question.

Questions 22 and 23 refer to the following table.

Unit	U.S. Equivalent	Metric Equivalent
1 ton	2,000 lb	0.907 metric ton
1 acre	43,560 sq ft	4,047 square m

22. What is the number of square feet in an acre, written in scientific notation?
- A. 0.4356×10^6
 B. 4.356×10^4
 C. 4.356×10^3
 D. 43.56×10^3

23. A shipment of goods weighs 5 tons. Which of the following expressions could be used to express the weight in metric tons?

- A. $5 \times 0.907 \times 10^{-1}$
 B. $5 \times 9.07 \times 10^1$
 C. $5 \times 9.07 \times 10^{-2}$
 D. $5 \times 9.07 \times 10^{-1}$

Answers and explanations begin on page 672.

Lesson 4: Order of Operations

When a mathematical expression contains more than one operation, its value may depend upon the order in which the operations are performed. To avoid confusion, mathematicians have agreed to perform operations in a certain order.

Key Ideas

- TO EVALUATE AN EXPRESSION CORRECTLY, YOU MUST FOLLOW THE ORDER OF OPERATIONS.
- IF AN EXPRESSION USES MORE THAN ONE SET OF GROUPING SYMBOLS, START WITH THE INSIDE SET AND WORK TO THE OUTSIDE.
- THE DIVISION BAR MAY BE USED AS A GROUPING SYMBOL.

The Order of Operations

1. Parentheses or any other grouping symbols that enclose operations
2. Exponents and roots
3. Multiplication and division, working from left to right
4. Addition and subtraction, working from left to right

Study the following example to see how to apply the order of operations. Notice that parentheses are used in two places in the expression; however, only the first set of parentheses encloses an operation.

Example 1: Evaluate the expression $\frac{(5+3)^2}{4} + 3(-1)$.

1. Perform the addition operation in parentheses. $\frac{(8)^2}{4} + 3(-1)$

2. Raise 8 to the second power. $\frac{64}{4} + 3(-1)$

3. Divide, then multiply. $16 + (-3)$

4. Add. 13

The value of the expression $\frac{(5+3)^2}{4} + 3(-1)$ is **13**.

In more complicated expressions, one set of grouping symbols may be nested within another set. To avoid confusion, you can also use brackets [] or braces { } to group operations. To evaluate an expression with more than one set of grouping symbols, work from the inside to the outside.

Example 2: Evaluate the expression $4[5(-4+3)+2]$.

1. Perform the operation in the inner set of grouping symbols: $(-4+3)$. $4[5(-4+3)+2]$
 $4[5(-1)+2]$

2. Do the operations inside the brackets. Since multiplication comes before addition in the order of operations, multiply 5 and -1 and then add 2. $4[-5+2]$
 $4[-3]$

3. Multiply 4 and -3 . -12

The division bar (also called a fraction bar) is also a grouping symbol. Before you divide, perform any operations shown above and below the bar.

GED® TEST TIP

You can use the parentheses keys on your online calculator to help you follow the order of operations.

Example 3: Evaluate the expression $\frac{15 + 25}{2(5)} + 6$.

1. Perform the operations above and below the fraction bar.

$$\frac{15 + 25}{2(5)} + 6$$

2. Divide, then add.

$$\frac{40}{10} + 6$$

$$4 + 6 = 10$$

Practice 4

A. Find the value of each expression. You MAY NOT use a calculator.

1. $4(3) - 2 + (6 + 4 \cdot 2)$

6. $\frac{25}{(4 + 1)} \cdot 3 + (6 - 1)$

11. $(4 + 2)^2 + (7 - 2)^3$

2. $16 \div (10 - 6)^2$

7. $2^3 + (8 - 5)^2 - 3$

12. $7^2 \div (11 - 4) + (9 + 14)$

3. $5^2 - (5 - 7)(2)$

8. $(4 - 12)(-6) + (10 - 3)$

13. $2 \left[(17 - 11)^2 \cdot \frac{(15 - 5)}{2} \right]$

4. $3(-3) + (7 + 4)$

9. $30 \div 3(5 - 4)$

14. $(5^2 + 6 - 3) \div (16 - 3^2)$

5. $\frac{3^3}{5 - 2} - \frac{(4 - 2)^2}{2}$

10. $15 + (4)(3) - 2^2$

15. $150 - 4 \left[\frac{3 + 9}{4 - 1} \cdot (14 - 11)^2 \right]$

B. Choose the one best answer to each question.

Question 16 refers to the following information.

Susan is in charge of planning Midvale Hospital's parent education classes. She uses the table below to determine the cost of each class to the hospital.

Midvale Hospital Parenting Workshops	
Type of Workshop	Cost per Participant
Childbirth Classes	\$35 per couple
Infant Care	\$30 per person
Teaching Your Child to Read	\$60 per person

16. A local foundation has offered to pay 75 percent of the cost of infant care classes. The hospital will cover any remaining costs. There are 28 parents enrolled in the upcoming class. Which of the following expressions could be used to find the amount, in dollars, the hospital will pay?

- A. $(75)(28)(30)$
- B. $(28)(30) - (0.75)(30)$
- C. $(1 - 0.75)(28)(30)$
- D. $(1 - 0.75)(30) + 28$

17. In the expression

$$5 + 2 \left[7 \left(\frac{10^2}{10} \right) + (6 - 2)(3) \right],$$

what is the last operation you should perform to find the value of the expression?

- A. Subtract 2 from 6.
- B. Add 5.
- C. Multiply by 2.
- D. Find the square of 10.

18. Find the value of the expression

$$22 + 6[(14 - 5) \div 3(17 - 14)].$$

- A. 2.73
- B. 28
- C. 76
- D. 97

Answers and explanations begin on page 672.

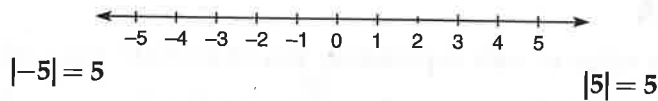
Lesson 5: Absolute Value

Key Ideas

- The magnitude of a number, without a sign, is called the absolute value.
- Absolute value is used to determine distance of a positive or negative number from zero.
- In the order of operations, absolute value is treated the same as a set of parentheses.

The **absolute value** of a number is its distance from zero on the number line. For example, the absolute value of 5 is 5, since 5 is five spaces away from zero on the number line. But the absolute value of -5 is also 5, since -5 is also five spaces away from zero on the number line. Because **distance** is always **positive**, the absolute value of any positive or negative number is always positive.

Absolute value is written as two straight lines around a number, like this: $|-5|$.



Example 1: Which of the following is equal to $|-13|$?

- (1) 13
- (2) -13

The correct answer is (1). Absolute value is always positive.

Treat the absolute value sign as you would treat parentheses in the order of operations. If numbers inside the absolute value sign are being added, subtracted, multiplied, or divided, then do that operation before you do anything else.

Example 2: Which of the following is equal to $-20|17 - 34|$?

- (1) 340
- (2) -340

The correct answer is (2). First perform the subtraction within the absolute value sign. Because you first found the absolute value, change -17 to 17. Then multiply by -20 , like this:

$$-20|17 - 34| = -20|-17| = -20(17) = -340.$$

We often think in terms of absolute value in real life, though we might not bother to call it that. For example, imagine that you have \$314 in a checking account but you write a check for \$400. You've overdrawn your account by \$86. The bank will record your balance as $-\$86.00$. But you'll think, "I'm \$86 short." You'll just think of the absolute value, or **magnitude**, of the overdraft, without the negative sign.

GED® TEST TIP

A signed number, positive or negative, is used to show its direction in relation to zero. However, when you see a number inside two straight lines, you are working with its distance from zero—which is always positive.

Practice 5

A. Find the absolute values.

- | | | | |
|-------------|--------------|-------------------|-----------------|
| 1. $ 18 $ | 4. $ 95 $ | 7. $ -5,708,432 $ | 10. $ 10.5 $ |
| 2. $ -107 $ | 5. $ -7026 $ | 8. $ -85.6 $ | 11. $ -163.24 $ |
| 3. $ 423 $ | 6. $ -18 $ | 9. $ 42 $ | 12. $ -3.14 $ |

B. Use absolute value to find the solutions.

- | | | |
|--------------------|-----------------------------|--------------------------|
| 13. $5 + -6 $ | 17. $- 110 - 201 $ | 21. $-5 -4 $ |
| 14. $-3 52 $ | 18. $-14 + -28 \div 2 $ | 22. $ -6 - -7 $ |
| 15. $3 -52 $ | 19. $706.2 - -86.4 + 0.2 $ | 23. $ 17 \div -8 $ |
| 16. $12 \div -4 $ | 20. $49 \div (- -7)$ | 24. $ -5.5 \times (-2)$ |

C. Choose the one best answer to each question.

25. The temperature in Northville at 9:00 p.m. was -5°F . By 5:00 a.m. the following morning, the temperature was -15°F . By how many degrees did the temperature change?
- A. -15 degrees
B. -5 degrees
C. 5 degrees
D. 10 degrees
26. Bob has errands to run. He walks 5 blocks east from his apartment to the barber shop, then walks 6 blocks west to the grocery store, then walks another 2 blocks west to the post office, and finally walks back home. Assuming that Bob's apartment, the barber shop, the grocery store, and the post office are all located on the same street, how many blocks did Bob walk in completing his errands?
- A. 6
B. 11
C. 13
D. 16
27. Milania has a score of -65 points, and Chris has a score of 55 points. By how many points is Milania losing to Chris?
- A. 55
B. 65
C. 120
D. 150
28. Absolute error is the absolute value of the difference between an actual value and its measurement. A deli scale gives a measurement of 25 ounces for a cut of meat that actually weighs only 23.5 ounces. What is the absolute error, in ounces, of the deli scale in this instance?
- A. 1.5
B. 23.5
C. 25
D. 48.5

Answers and explanations begin on page 672.